SHMM P1 Solutions

Allen Tang, Alex Kong

April 2025

1 In a triangle with integer side lengths, two sides have length 8 and 13. What is the absolute difference between the largest and smallest possible side length for the third side?

Answer: 14
The maximum side length has to be less than the sum of the two other side lengths: $8 + 13 - 1 = 20$
The minimum side length has to be greater than the absolute difference of the two other sides: $13 - 8 + 1 = 6$
Therefore the absolute difference is $20 - 6 = 14$

2 Several squares are drawn on a line with length 10. What is the area of the sum of all the squares?

Answer: N/A
Sorry I wrote the question wrong.

3 How many ways are there to rearrange the letters in SAGE? It does not have to be a real word: for example EGAS is a possible rearrangement.

Answer: 24
Consider the four letter word to be four empty slots that each need to have a letter (S,A,G,E).
There are 4 choices for which letter goes in the first slot. We cannot use that letter again, so the second slot has 3 possible letters. Then, the third slot has 2

possible letters and the last slot has only 1. Thus, we multiply $4*3*2*1 = 24$ ways.
(note: we would also accept 23, as technically, SAGE is not a rearrangement of itself, so we would have to subtract 1)

4 What is the value of 2(2(2(2(2+1)+1)+1)+1)+1?

Answer: 63
Notice that the result is $2^5 + 2^4 + 2^3 + 2^2 + 2^1 + 2^0 = 2^6 - 1 = 63$

5 Ms. Dear is going on a roadtrip across the United States. Her odometer currently reads 12345 miles. How many miles does she drive until her odometer reads a palindrome? A palindrome is a number that reads the same forwards and backwards. For example, 808 and 1331 are palindromes.

Answer: 76
The next palindrome will be 12421. Thus, we do 12421-12345=76.

6 Quadrilateral ABCD is a rhombus with perimeter 52. Diagonal AC=24. What is the area of rhombus ABCD?

Answer: 120
Let the intersection of the two diagonals of the rhombus be O. Notice that the side length of the rhombus is $\frac{52}{4} = 13$ and $OA = OC = \frac{24}{2}$. Notice the right triangles formed by the two diagonals and the sides, ex: ΔABO. The hypotenuse of this triangle is $AB = 13$, one of the sides is $OA = 12$, and therefore the other side is $OB = \sqrt{13^2 - 12^2} = 5$.
Notice the area of the rhombus is $AC * OB = 24 * 5 = 120$

7 In the battery collection contest, the number of batteries the sophomores collected was 150 more than 150 percent the number of batteries the freshmen collected. The number of batteries the juniors collected was 20 less than 20 percent of batteries the freshmen collected. The number of batteries the seniors collected was 200 more than 200 percent the number of batteries the sophomores collected. In total, the four grades collected 1200 batteries. How many batteries did the juniors collect?

Answer: 0
Let the number of batteries the freshmen collected be x. We can now write:
Freshies: x
Sophomores: $1.5x + 150$
Juniors: $0.2x - 20$
Seniors: $2(1.5x + 150) + 200 = 3x + 500$
Now, we can add these all up to equal 1200 total batteries:
$5.7x + 630 = 1200$
So $x = 100$
Thus, the juniors collected $0.2(100) - 20 = 0$ batteries.

8 Mr. Lau's bag contains 2 triangles, 4 squares, and 6 pentagons. He draws 3 shapes at random from the bag without replacement. What is the probability that all 3 shapes are of different types?

Answer: $\frac{12}{55}$
There are 3! different ways to order the 3 shapes chosen. Then we multiply by the probability without replacement of choosing three different shapes:
$3! * \frac{6}{12} * \frac{3}{11} * \frac{2}{10} = 6 * \frac{1}{2} * \frac{3}{11} * \frac{1}{5} = \frac{12}{55}$

9 9 students took Mr. Schumacher's AP World test on Friday. Their scores are the elements of the set $S = 39, 46, 43, 42, 42, 45, 36, 49, 45$. Then, on Monday, 2 of the students who were sick then took the test. Once Mr. Schumacher added their scores to S the mean score increased by 1, the median didn't change, and there emerged a unique mode. What is the absolute difference of the scores of the 2 students who took the test on Monday?

Answer: 13
We start by ordering S from least to greatest:
$36, 39, 42, 42, 43, 45, 45, 46, 49$
Currently, the total sum of these scores is 387. The mean is $\frac{387}{9} = 43$. After the 2 sick students take the test, the mean will become 44, so the total sum must become $44 * 11 = 484$. Thus, the sum of the two scores of the 2 students is $484 - 387 = 97$.
Currently, the modes of the original scores are 42 and 45. So, for the addition of the scores of the 2 new students is to create a unique mode, there are only 2 possibilities:
Either the mode is 42 or 45.
Any other mode doesn't work because the two scores can't be the same because they have to add up to 97.
So let's try for one of the scores to be 42:
Then the other score will be $97 - 42 = 55$ which works because it doesn't change the median.
Otherwise if one of the scores is 45:
The other score is $97 - 45 = 52$ which doesn't work because it changes the median of the set to 45.
Therefore the two test scores are 42 and 55. Their absolute difference is $55 - 42 = 13$

10 Francis Frog chooses a nonnegative integer less than ten-thousand, completely at random. Feodor Frog chooses a four-digit positive integer whose digits are all between 2 and 5; inclusive, also completely at random. What is the probability that the Francis Frog's number is greater than Feodor Frog's number?

Answer: $\frac{12221}{20000}$
The average value of Feodor Frog's number is $\frac{2+3+4+5}{4} * 1000 + \frac{2+3+4+5}{4} * 100 + \frac{2+3+4+5}{4} * 10 + \frac{2+3+4+5}{4} = 3888.5$
We want Francis Frog's number to be greater than Feodor Frog's number's average value, meaning that it needs to be greater than 3888.5.
There are 10000 integers in total from 0 to 9999, and there are 3889.5 "integers" from 0 to 3888.5.
So the probability that Francis Frog's number is greater than Feodor Frog's number is $\frac{10000-3889.5}{10000}$ which simplifies into $\frac{12221}{20000}$